

App biking

Max Pérez Sastre

Avís legal

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1. introduccio

2. objectis

3. Metodologia

En aquest treball he utilitzat diferents eines i tecnologies que m’han permès entendre millor com es desenvolupa un projecte complet de programació. La meua manera de treballar ha estat sobretot **aprendre fent**: investigar, provar, equivocar-me, corregir i finalment aconseguir un resultat funcional. Aquest procés m’ha ajudat no només a obtenir un programa capaç d’integrar les dades de Bicing i AMBici, sinó també a descobrir i aprendre conceptes nous relacionats amb la programació, el tractament de dades, el control de versions i la documentació tècnica.

La metodologia no ha estat lineal, sinó que ha combinat moments de recerca amb proves pràctiques. Moltes vegades he hagut de buscar informació sobre com funcionen les APIs, altres cops he hagut de provar diferents maneres de calcular distàncies o de filtrar dades, i també he hagut d’investigar com treballar amb Git, amb màquines virtuals o amb LaTeX per a la documentació. Això m’ha permès entendre que en informàtica és tan important saber resoldre problemes com saber on buscar la informació adequada. Per explicar tot aquest procés, he dividit la metodologia en diversos blocs. En primer lloc, parlaré de l’ús de **Python** com a llenguatge de programació i de com he après a estructurar el codi i treballar amb dades en temps real. A continuació, explicaré com he treballat amb les **APIs** de Bicing i AMBici, i com he resolt el repte de calcular **distàncies i punts de transbord**. També dedicaré un apartat al control de versions amb **Git**, a la configuració d’una **màquina virtual** per garantir la reproduïbilitat i, finalment, a l’ús de **LaTeX** i **Kile** per redactar aquesta memòria de manera professional. El capítol es tancarà amb una síntesi dels **aprenentatges generals** adquirits durant tot el projecte.

3.1 Eines utilitzades

3.1.1 Màquina virtual i entorn de treball

Per desenvolupar aquest projecte he decidit utilitzar una **màquina virtual** com a entorn principal de treball. Una màquina virtual és un programa que permet executar un sistema operatiu complet dins d’un altre ordinador. És com tenir un ordinador dins

de l'ordinador: amb el seu propi disc dur, memòria i configuració, però compartint el maquinari físic de la màquina principal.

He triat treballar amb una màquina virtual per diversos motius. En primer lloc, em permetia tenir un entorn **net i controlat**, sense risc de fer malbé el sistema principal amb instal·lacions de programes o configuracions equivocades. En segon lloc, m'assegurava que el projecte es podia reproduir fàcilment en altres ordinadors, ja que sempre tindria el mateix sistema operatiu i les mateixes versions de Python i les llibreries necessàries. Finalment, m'ha ajudat a entendre millor conceptes bàsics de la informàtica moderna, com la **virtualització** i l'aïllament d'entorns.

La màquina virtual que vaig configurar estava basada en Linux, ja que és un sistema operatiu molt utilitzat en programació i administració de sistemes. Dins d'aquest entorn vaig instal·lar totes les eines necessàries: Python, el gestor de paquets **pip**, Git i l'editor Kile per a LaTeX. Així, tenia tot el projecte centralitzat en un entorn específicament preparat per a ell.

Gràcies a aquesta decisió he après que utilitzar màquines virtuals no només és una qüestió tècnica, sinó també una bona pràctica professional. En equips de treball grans, cada programador pot tenir una màquina virtual idèntica a la dels seus companys, i això evita molts problemes de compatibilitat. També he entès que aquesta és la base de moltes tecnologies actuals, com els contenidors Docker o els serveis al núvol.

En resum, l'ús de la màquina virtual m'ha permès treballar amb més seguretat i confiança, assegurant-me que el meu projecte era **reproductible, portable i professional**.

3.1.2 Ús de Python per al desenvolupament

El llenguatge de programació que he triat per aquest projecte és **Python**. Python és actualment un dels llenguatges més utilitzats arreu del món, especialment en camps com la ciència de dades, la intel·ligència artificial i la programació acadèmica. Una de les seves grans avantatges és la seva **senzillesa**, ja que té una sintaxi clara i fàcil d'entendre, però alhora disposa d'una enorme quantitat de llibreries que permeten resoldre problemes complexos.

En el meu cas, Python ha estat l'eina ideal per treballar amb **dades en temps real** provinents de dues fonts diferents: l'API del Bicing i l'API d'AMBici. Amb la llibreria

`requests` he pogut descarregar la informació en format **JSON**, i posteriorment l'he processat per obtenir només els camps necessaris: nom de l'estació, coordenades geogràfiques, bicicletes lliures i espais disponibles. També he afegit un indicador nou, anomenat **canvi ràpid**, que calcula quants intercanvis immediats es poden fer en una estació (és a dir, el mínim entre bicis lliures i espais lliures).

Un altre aspecte important ha estat l'**organització del codi**. He après a dividir el programa en **funcions** independents, cadascuna amb una responsabilitat clara: obtenir dades, calcular distàncies, filtrar estacions, trobar punts de transbord, etc. Aquesta manera de treballar fa que el programa sigui més fàcil d'entendre i de mantenir. També he fet servir la llibreria `argparse` per afegir paràmetres a la línia de comandes, de manera que l'usuari pot escollir quines opcions vol executar, com ara `--top`, `--barri`, `--coords` o `--transbord`.

Finalment, he incorporat la possibilitat d'**exportar dades en CSV**, un format molt comú que permet obrir els resultats en programes com Excel. Això m'ha ensenyat que un programa no només ha de mostrar informació a la pantalla, sinó també oferir opcions per **guardar i reutilitzar** aquestes dades.

En resum, gràcies a Python he pogut entendre millor com es treballa amb dades reals, com es dissenya un programa modular i flexible, i com es poden crear eines pràctiques que resolguin necessitats concretes.

3.1.3 Treball amb APIs i dades en temps real

Una de les parts més importants d'aquest projecte ha estat aprendre a utilitzar **APIs** per obtenir dades en temps real. Una API (Application Programming Interface) és un servei que permet connectar-se a una aplicació i obtenir informació o enviar-li instruccions. En el cas de Bicing i AMBici, les APIs ofereixen informació constantment actualitzada sobre l'estat de les estacions i les bicicletes disponibles.

Per al **Bicing** vaig utilitzar el servei de **CityBikes**, que recull informació de moltes xarxes de bicicletes compartides d'arreu del món. Aquest servei em permet descarregar totes les estacions de Bicing de Barcelona en format **JSON**. JSON (JavaScript Object Notation) és un format molt utilitzat a Internet per intercanviar dades: és lleuger, fàcil d'entendre i pot ser llegit directament per Python com si fossin diccionaris i llistes.

Per a l'**AMBici** la situació era diferent. No hi havia una API tan coneguda com la de Bicing, i vaig haver d'investigar fins a descobrir que funciona amb l'estàndard **G-BFS (General Bikeshare Feed Specification)** a través de la plataforma Nextbike. Aquest estàndard defineix diferents arxius JSON: un amb la informació fixa de les estacions (nom, coordenades, capacitat), i un altre amb l'estat en temps real (bicis lliures i espais lliures). Això em va ensenyar que cada servei pot publicar la informació de manera diferent i que cal adaptar el codi per combinar-los en un sol programa.

Treballar amb aquestes APIs m'ha obligat a aprendre a **analitzar la documentació** per entendre què significa cada camp del JSON i com s'ha d'interpretar. També he après a gestionar situacions en què l'API pot fallar: potser la connexió no respon, o el format de dades canvia lleugerament. Per això he afegit al codi comprovacions i missatges d'error que ajuden l'usuari a entendre què ha passat.

En resum, gràcies al treball amb APIs i JSON he après que tenir dades no és suficient: cal entendre-les, filtrar-les i transformar-les en informació útil. Aquesta part ha estat molt enriquidora perquè m'ha ensenyat a connectar-me amb serveis externs i a integrar-los en un programa propi.

3.1.4 Càlcul de distàncies i punts de transbord

Per poder determinar si dues estacions estan a prop, he hagut d'aprendre a calcular distàncies a partir de les coordenades de latitud i longitud. Com que la Terra és gairebé esfèrica, no es poden calcular les distàncies simplement amb fórmules planes com el teorema de Pitàgores. Per això he utilitzat la **fórmula de Haversine**, molt emprada en geografia i navegació.

Aquesta fórmula té en compte la curvatura de la Terra i permet obtenir la distància entre dos punts a partir de les seves coordenades. En aquest projecte s'ha utilitzat per dues funcionalitats concretes:

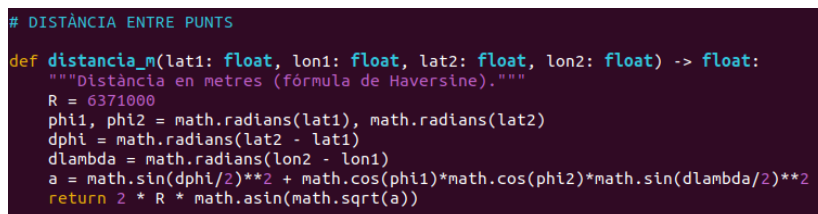
- Filtrar les estacions dins d'un **radi** determinat a partir d'unes coordenades concretes (opció `--coords`).
- Detectar **punts de transbord** entre Bicing i AMBici, és a dir, estacions de dues xarxes situades a menys de 250 metres.

La implementació en Python d'aquesta fórmula al meu programa és la següent:

```
def haversine(lat1, lon1, lat2, lon2):
    R = 6371000 # radi de la Terra en metres
    phi1, phi2 = radians(lat1), radians(lat2)
    dphi = radians(lat2 - lat1)
    dlamba = radians(lon2 - lon1)

    a = sin(dphi/2)**2 + cos(phi1)*cos(phi2)*sin(dlamba/2)**2
    c = 2 * atan2(sqrt(a), sqrt(1-a))

    return R * c
```



```
# DISTÀNCIA ENTRE PUNTS
def distancia_m(lat1: float, lon1: float, lat2: float, lon2: float) -> float:
    """Distància en metres (fórmula de Haversine)."""
    R = 6371000
    phi1, phi2 = math.radians(lat1), math.radians(lat2)
    dphi = math.radians(lat2 - lat1)
    dlamba = math.radians(lon2 - lon1)
    a = math.sin(dphi/2)**2 + math.cos(phi1)*math.cos(phi2)*math.sin(dlamba/2)**2
    return 2 * R * math.asin(math.sqrt(a))
```

Figure 3.1: Funció Haversine implementada al programa.

3.1.5 Git i control de versions

Una altra eina fonamental en aquest projecte ha estat l'ús de **Git** com a sistema de control de versions. Git és una eina que permet enregistrar tots els canvis que es fan en un projecte de programació. Això vol dir que cada modificació del codi queda guardada en un historial, i és possible tornar enrere si alguna cosa falla, o comparar diferents versions del programa.

Vaig decidir utilitzar Git perquè és una pràctica habitual en el món professional i perquè em donava més seguretat al treballar. En lloc de guardar moltes còpies del fitxer amb noms diferents (per exemple, `programa_final_v3.py`), amb Git puc tenir un repositori únic i un registre clar de cada canvi que faig.

Al llarg del projecte he après a utilitzar les comandes més importants:

- `git status` per comprovar l'estat del repositori i veure quins fitxers han canviat.

- `git add` per preparar els fitxers que vull guardar.
- `git commit-a` per enregistrar els canvis amb una descripció.
- `git push` per pujar el repositori a GitHub i tenir-ne una còpia en línia.
- `git pull` per baixar possibles canvis del repositori remot i sincronitzar-lo amb la meua còpia local.

A més, he creat un repositori a **GitHub**, una plataforma molt utilitzada per compartir codi. Això em permet no només tenir una còpia de seguretat del meu projecte, sinó també entendre com es podria treballar de manera col·laborativa amb altres programadors.

El treball amb Git m'ha ensenyat que el control de versions és molt més que una eina tècnica: és una manera d'organitzar el projecte i de treballar de manera **professional**. En resum, gràcies a Git he après la importància de registrar els canvis, documentar-los i tenir el codi sempre disponible i protegit.

3.1.6 Conclusions i aprenentatges generals

Aquest projecte m'ha servit no només per crear un programa funcional, sinó sobretot per adquirir una gran quantitat de coneixements i habilitats que abans no tenia. Al principi em semblava que es tractava només de descarregar dades de dues plataformes i mostrar-les a l'usuari, però he descobert que darrere hi havia molts reptes tècnics i conceptuals.

Pel que fa a la part tècnica, he après a:

- Programar de manera més estructurada amb Python, utilitzant funcions, arguments i opcions a la línia de comandes.
- Treballar amb **APIs** i entendre el format **JSON**, que és la base de la comunicació entre serveis d'Internet.
- Aplicar fórmules matemàtiques reals, com la de **Haversine**, per resoldre problemes pràctics en programació.

- Controlar les versions del projecte amb **Git** i aprendre bones pràctiques professionals de desenvolupament.
- Utilitzar una **màquina virtual** per tenir un entorn segur, net i reproducible.
- Redactar documentació amb **LaTeX i Kile**, aconseguint un resultat final molt més professional del que hauria estat amb un processador de text convencional.

Més enllà de la tècnica, aquest treball m'ha ensenyat també una manera diferent de pensar i d'organitzar-me. He après a investigar per la meva compta, a buscar documentació, a provar i equivocar-me, i sobretot a ser constant fins aconseguir el resultat esperat. Aquest procés m'ha donat confiança en les meves capacitats i m'ha fet adonar que puc afrontar reptes complexos si els divideixo en parts i avanço pas a pas.

En conclusió, aquest projecte ha estat una experiència molt enriquidora. He après a programar millor, a utilitzar eines noves i a entendre millor el món de les dades i els serveis digitals. Però, sobretot, he après que la programació no és només escriure codi: és també **resoldre problemes, comunicar resultats i aprendre contínuament**.

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